STATEMENT OF PURPOSE

Applicant: Xuan Thanh Trinh
Program: Ph.D. in Electrical, Electronics and Computer Engineering
Proposed Advisor: Assistant Professor Van-Hai Bui
Institution: University of Michigan-Dearborn

Introduction

As a final-year Electrical Engineering student at Hanoi University of Science and Technology (HUST), my research at the Power Grid and Renewable Energy (PGRE) Laboratory focuses on developing distributed mathematical optimization models for decentralized energy management. While this foundation drove my graduation thesis on microgrid resilience, I recognized that classical optimization alone cannot scale to the real-time computational demands of modern power systems. To overcome these computational bottlenecks, I require advanced training in scalable, cyber-physical control systems. The Ph.D. program at UM-Dearborn directly addresses this need, providing the exact framework necessary to become a highly specialized research engineer capable of designing decentralized architectures for high renewable penetrations.

Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering and Renewable Energy instilled in me a systems-level perspective. Rather than isolating individual grid components, I trained to deconstruct complex power networks into dynamic, tractable optimization problems. This mathematical foundation drives my independent research trajectory, allowing me to architect solutions that bridge abstract algorithms with real world physical constraints.

This trajectory reflects a systematic transition from treating power grids as aggregate generation-load balances to modeling them as complex computational graphs. In research recently submitted to *Sustainable Energy, Grids and Networks* (SEGAN), I established and validated a peer-to-peer trading framework that subdivides large-scale distribution networks into autonomous microgrids, integrating rigorous grid physics (AC-OPF) with distributed optimization (ADMM) across both radial and ring-connected topologies. By accounting for marginal transmission losses, this architecture reduced operational costs while strictly bounding voltage levels within the $\pm 5\%$ threshold, preventing local trading from destabilizing the wider public grid. However, this static framework lacked the temporal foresight to handle sudden local failures.

To address this vulnerability, my graduation thesis was designed to enhance grid resilience and achieve dynamic fault tolerance. Specifically, I layered Model Predictive Control (MPC) over the trading baseline to coordinate emergency power exchanges and backup diesel generation. During simulated localized outages, this temporal coordination empowered neighboring networks to assist a failing grid, reducing forced load shedding from 15 MWh down to just 3 MWh while protecting 100% of critical loads.

Ultimately, resolving complex convergence errors has built my persistence in strict code debugging. Beyond independent problem-solving, I effectively coordinate with multiple advisors and research teams. Equipped with a robust technical toolkit (Python, Pyomo), I am ready to deploy this foundation within your ecosystem to drive scalable, real-time grid operations.

Research Interests

My academic trajectory is driven by two primary research directions:

Currently, I focus on ensuring microgrid resilience through mathematical optimization. My theoretical power systems knowledge provides the foundation to accurately model physical grid constraints, while optimization techniques like MILP, SOCP, and MPC serve as the practical bridge to solve complex operational challenges.

My future research expands into advanced artificial intelligence, specifically Multi-agent Reinforcement Learning, for autonomous energy management. I aim to leverage these learning-based policies as computational accelerators, enabling real-time dispatch across decentralized grids. Crucially, this AI-driven framework provides a robust mechanism to overcome modeling inaccuracies and strict mathematical assumptions in classical solvers.

Ultimately, rather than pursuing these domains in isolation, I intend to integrate them into a mutually reinforcing architecture. My overarching objective is to leverage this synergy between physics-based modeling and data-driven learning to architect certifiably safe, autonomous power systems capable of adapting to highly dynamic and unpredictable grid conditions.

Academic and Research Alignment

The University of Michigan-Dearborn provides the rigorous theoretical coursework and collaborative research environment required to pursue my synergistic vision. Specifically, my trajectory aligns directly with the laboratory of Prof. Van-Hai Bui, driven by a shared focus on autonomous multi-agent energy management. Building upon the deterministic optimization foundation detailed previously, I aim to contribute to his learning-based frameworks, such as FairMarket-RL and XRL-LLM, by embedding my mathematical models as physics-informed safety filters.

Beyond Prof. Bui's laboratory, the expertise of Dr. Wencong Su and Dr. Junho Hong strongly complements my holistic research vision. Dr. Su's research on utilizing LLMs for Economic Dispatch would deepen my understanding of system-level AI integration and large-scale energy management. Conversely, Dr. Hong's focus on physics-informed anomaly detection would strengthen my foundation in component-level physical constraints and grid security. Fusing these perspectives expands my research horizon, providing the deep insights necessary to engineer autonomous grid architectures that dynamically scale through real-time volatility while preserving strict physical safety guarantees.

Career Objectives

Following the completion of my Ph.D., I intend to pursue a career as an R&D scientist dedicated to the advancement of global power infrastructures. My long-term vision is to translate theoretical research into scalable, real-world energy solutions. Furthermore, I am deeply committed to applying these breakthroughs to solve high-impact challenges in developing regions, specifically contributing to the technical modernization of Vietnam's national grid. Ultimately, the University of Michigan-Dearborn provides the ideal collaborative environment where I can contribute my foundational expertise to the department's ongoing innovations while leveraging its academic rigor to realize these global solutions.